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台灣地震科學中心簡訊

Taiwan Earthquake Research Center Newsletter

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台灣地震科學資料中心

The Taiwan Earthquake Research Data Center (TECDC)

台灣地震科學中心地震儀器服務平台

Taiwan Earthquake Research Center Instrument Service

台灣地震科學中心地球物理儀器服務平台

The Geophysical Instrument Service

本季成員發表文章 (2022.03~)

致謝

前言

郭陳澔 台灣大學地質科學系 教授
台灣地震科學中心 執行秘書

TEC 簡訊也有 DOI 了！為了讓 TEC 簡訊中的文章能夠適當被引用，從今年開始，在 TEC 簡訊中的每篇文章都有專屬的 DOI，並溯及過去所有的文章。此次簡訊邀請清華大學游能悌教授介紹近 2400 年來在台灣北部地層中的海嘯紀錄，同時並邀請 TEC 三大服務平台（資料、地震與地球物理）的計畫主持人撰寫現階段各服務平台的現況與成果，希望學界在享有平台服務的同時，能在文章發表上適當地在致謝中引用，讓平台的服務能夠被看見。

儀器服務平台

- 地震儀器中心儀器庫存查詢系統目前更新中，預計新版網頁將於 111 年 9 月底上線，若您有儀器使用之需求請與儀器中心聯絡。
- TEC 支援各大專院校相關科系之地球物理實習課程。歡迎有需求的授課教師於線上系統提出申請，申請網頁為：https://tec.earth.sinica.edu.tw/geoclass/

近期消息

- **第六屆台灣地震科學中心年會**，預計於 111 年 9 月 21 日至 23 日假澎湖澎澄飯店舉辦，議程初步排定一天半會議及半天野外行程，會議相關資訊將盡快公告，歡迎踴躍參加。
- **2022 台日紐地震災害研討會 (2022 Taiwan-Japan-New Zealand Workshop on Earthquake Hazard)**，預計於 111 年 10 月 30 日至 11 月 4 日假台東鹿鳴溫泉酒店舉辦，會議相關資訊將盡快公告，歡迎踴躍參加。

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台灣北部長達 2400 年的西太平洋海嘯與浮石筏紀錄：海嘯再現間隔、浮石來源與漂流路徑

游能悌 國立清華大學通識教育中心
doi: 10.30067/TECNL.202206_1(1).0001

台灣位於聚合板塊邊界環繞的西太平洋 (菲律賓海)，常有劇烈的地震與猛爆的矽質火山噴發，形成能穿透大洋的災害性海嘯與浮石筏。從十九世紀起，北海岸就有一次大海嘯 (1867 基隆) 與一次浮石堆積 (1924) 的歷史紀錄，但是苦無更早的文獻，因此海嘯再現間隔、浮石來源與漂流路徑等重要資訊，付之闕如。

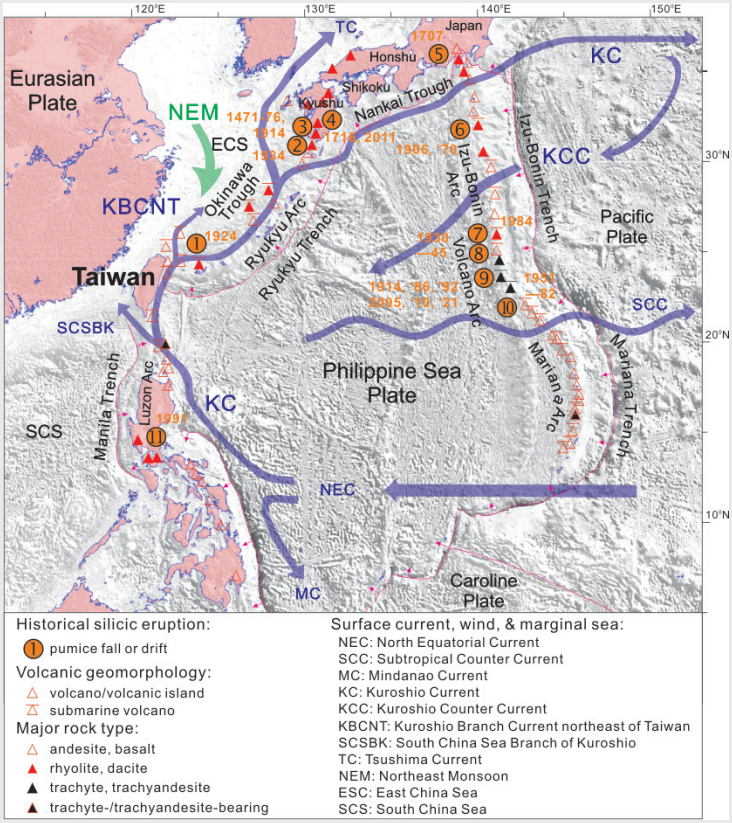
從 2013 年起，我們在金山與基隆的陸相氾濫平原岩芯與岩壁崩積層中，發現 7 層海相事件堆積層，年代不老於 2400 年。堆積層分布達內陸約 0.8 公里、高程在 2.4 到 11.5 公尺，超過當地颱風溢淹與暴潮所及，因此歸因於海嘯 (Yu et al., 2022)。較早的 1400 年中有 3 次海嘯，平均間隔是 472±39 年。較近的 1000 年中有 4 次海嘯，平均間隔只有 195±45 年，包括 1867 年基隆地震海嘯，與疑似對應 1694 年康熙台北湖地震的海嘯 (Sugawara et al., 2019; Yu et al., 2016; Yu et al., 2020)。

另外，黑色與白色浮石大量出現在 6 層海嘯堆積層中 (圖一)。黑色浮石的化學組成類似粗面岩與粗面安山岩，可能來自 2000 公里外的小笠原島弧南部 (圖二；Izu-Bonin)，藉由黑潮反流 (Kuroshio Counter Current) 向西到琉球，再跟著台灣東北黑潮支流 (Kuroshio Branch Current in NE Taiwan) 向北到東海，最後隨著東北季風與向岸波南下抵達北海岸。在 2021 年 8 月，小笠原島



圖一、金山海岸岩壁崩積層中的海嘯堆積礫石層。S 代表沈積岩礫石。未標示的礫石都是浮石。黑色浮石容易氧化，呈現橘紅的鐵銹色；白色浮石不易破裂，顏色偏淡黃。

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圖二、西太平洋的板塊、活火山與表面洋流。

弧南部的福德岡之場海底火山也爆發了，浮石筏約 2 個多月就抵達台灣。白色浮石組成類似低鉀的流紋岩與英安岩，也可能來自 2000 公里外的小笠原島弧南部與馬里亞納島弧北部，也可能來自 200 公里外的琉球島弧南部，就像 1924 年的浮石筏來自西表島北部的海底噴發。

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台灣地震科學資料中心

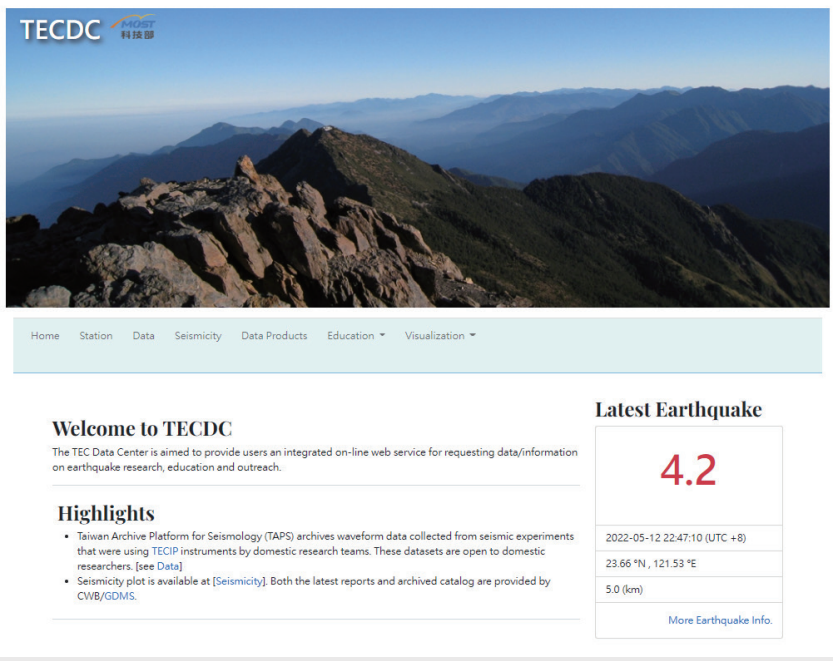
梁文宗 中央研究院地球科學研究所
doi: 10.30067/TECNL.202206_1(1).0002

摘要

台灣地震科學資料中心 (TECDC) 由科技部資助，主要目的是促進台灣地震科學學界的資料資源整合、開發即時地震學系統，並建置成果平台分享學界的研究成果，作為其他相關研究的進一步參考；除此之外，本團隊也同時開發地震科學教育推廣的線上活動，為培育下一代的科研人才盡一份心力。TECDC 的網址為：https://tecdc.earth.sinica.edu.tw (圖一)。

資料整合

在資料整合的工作上，我們首先重整了中研院地球所負責營運並對外開放的「台灣寬頻地震觀測網 (Broadband Array in Taiwan for Seismology, BATS)」網路服務系統，以符合現代地震觀測研究需求的網路技術與標準，使用者僅須填入所需的測站、分量、時段與輸出格式等資訊，即可點擊連結進行資料處理並下載，甚至可在工作站的命令列直接索取資料。這個資料服務的網址在：https://bats.earth.sinica.edu.tw/。根據台灣地震科學中心 (TEC) 的儀器與資料管理辦法，凡是向台灣地震科學中心儀器中心 (TECIP) 借用儀器所繳回的地震波形，都必須在實驗計畫結束後兩年內繳回原始資料到資料中心，並於計畫結束兩年後對國內學者開放，而為了協助各團隊以高效能方式處理資料並建置標準資料庫，TECDC 利用中研院地球所所採購的 Antelope 軟體套件，開設訓練研習營，建成標準資料庫之後可進一步進行自動化地震定位，讓地震觀測的工作可以順利展開，也預備好可供開放的標準波形資料庫；據此，我們在掌握相關的技術與標準之後，打造全新的「台灣地震科學資料服務平台 (Taiwan Archive Platform for Seismology, TAPS)」，這個服務平台的特色是可以只用一組帳號查詢下載所有過去觀測計畫所繳回的資料，大幅縮減逐一查詢各計畫資料庫的時間，也讓同仁們在發表成果時有資料網頁可提供讀者參考，平台網址為：https://taps.earth.sinica.edu.tw。TECDC 團隊所累積的經驗與技術也用來與氣象局合作建置了「台灣地震與地球物理資料管理系統 (GDMS) 」，將氣象局各個觀測網歷年來所收集的觀測數據開放給全球各界使用：https://gdmsn.cwb.gov.tw。圖二是 TECDC 收集台灣地區幾個觀測網詮釋資料在台灣西南部的測站分布。



圖一、台灣地震科學資料中心 (TECDC) 的入口網站首頁。

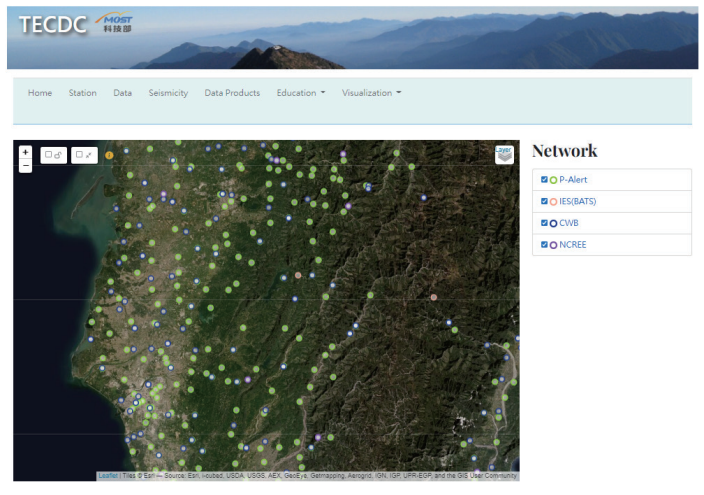
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即時地震學

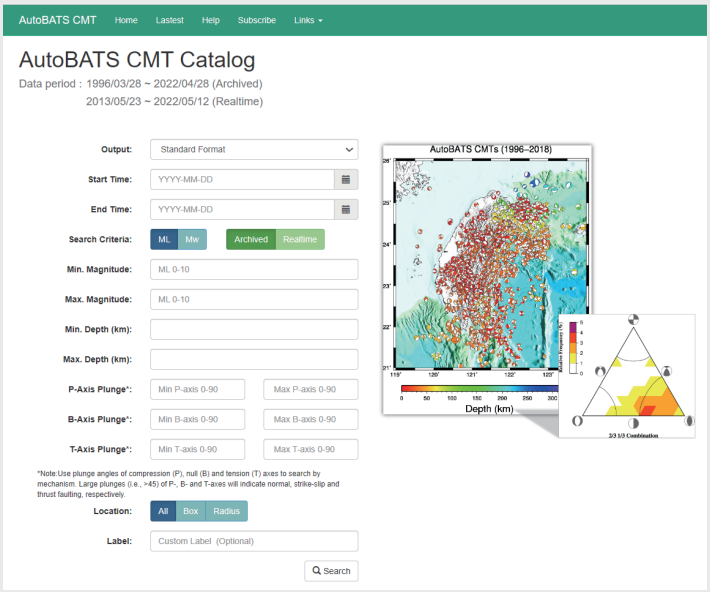
TECDC 在即時地震學的實務操作上與各團隊合作進行研發，例如目前由 TEC 維護的 P 波警報器強震網 (P-Alert Strong Motion Network) 近即時震度與地震事件波形資料庫 (Wu et al., 2013) (https://palert.earth.sinica.edu.tw)，而利用寬頻波形資料反演震源機制的解算系統除了與李憲忠博士合作的即時地震矩張量監測系統 (Lee et al., 2013; http://rmt.earth.sinica.edu.tw) 之外，也改良了 BATS 的地震矩張量解速報系統並全面自動化，可在地震後 5 分鐘自動在台灣地震科學資訊系統 (http://tesis.earth.sinica.edu.tw) 發布震源解的資訊 (AutoBATS 震源機制解 , Jian et al., 2018)，或可直接到 AutoBATS 的資料庫網站查詢瀏覽，網址為 https://tecdc.earth.sinica.edu.tw/FM/AutoBATS (圖三)，自動解再經人工判讀確認後將透過電子郵件通知訂閱的使用者，歡迎各界訂閱利用。此外，在大規模的強震發生時，鄰近的測站可能出現波形飽和的問題，且點源的假設恐不利破裂複雜震源解算，所以我們採用與周邊區域的觀測網合作的方式，即時取得較遠距離的測站資料，並利用 W-Phase 震源反演的技術 (WP, Kanamori and Rivera, 2008)，在接收氣象局的速報之後，即刻啟動區域 WP 反演以決定更可靠的地震矩規模與震源參數，這項成果展示在 https://tecdc.earth.sinica.edu.tw/FM/WP/，目前可解算的最小地震矩規模約為 4.4，和測站的包覆性有關。

國內研究成果平台

我們針對運用地震資料所研發的學術成果進行數值管理與視覺化服務，第一個成果是台灣速度模型 (Taiwan Velocity Model, TVM) 計畫，這個展示平台收集自 1995 年以來已在期刊上發表的三維速度與衰減構造模型，並參考 IRIS 的 EMC 網站設計 (IRIS DMC, 2011)，使用者可在各模型資料的涵括範圍內任取一剖面，然後選擇參數繪出對應的二維剖面速度或衰減構造，另可套疊地震活動或震源機制，繪製的成果圖及對應的數據均可直接下載，TVM 的展示網址為 https://tecdc.earth.sinica.edu.tw/TWtomo/ (圖四)，最新的速度模型 (Chen et al., 2019) 即將納入。除此之外，本團隊連接氣象局地震中心的即時測報成果，以線上繪圖的方式快速展示地震的時空分布，做為面對顯著地震時快速瀏覽地震活動 (Taiwan Seismicity Map) 時空分布的依據。



圖二、台灣地區不同單位負責的地震觀測網在西南部的測站分布。

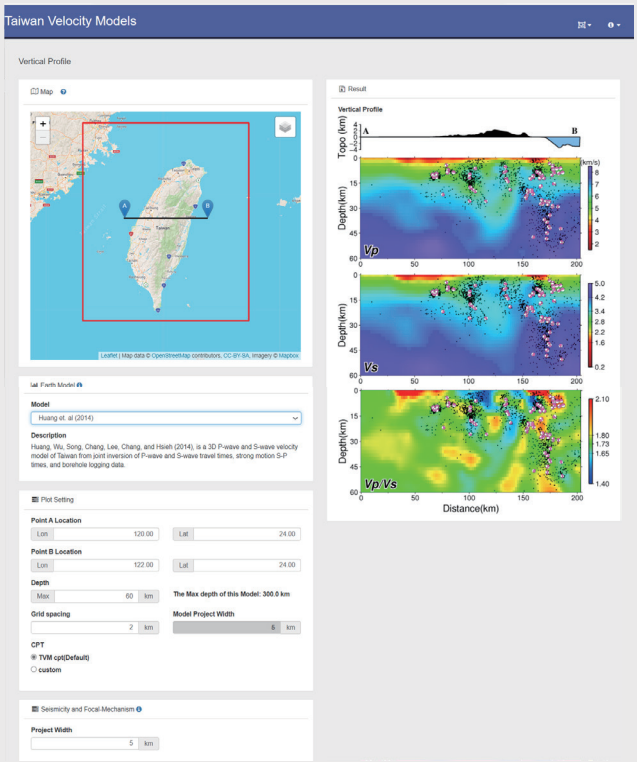


圖三、AutoBATS 自動震源機制解的資料庫首頁。

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地震科學教育推廣

過去本團隊陸續開發一些線上的教室活動，提供民眾自學或教學使用 (Liang et al., 2016)，也製作簡易型的地震儀 (TEC-100) 製做教學材料，甚至是台灣地區三維地震活動的紙模型 (圖五)，推廣台灣的地震科學成果與地震相關常識，這些成果都放在 TECDC 的教育 (Education) 功能連結。



圖四、台灣速度模型的展示平台。

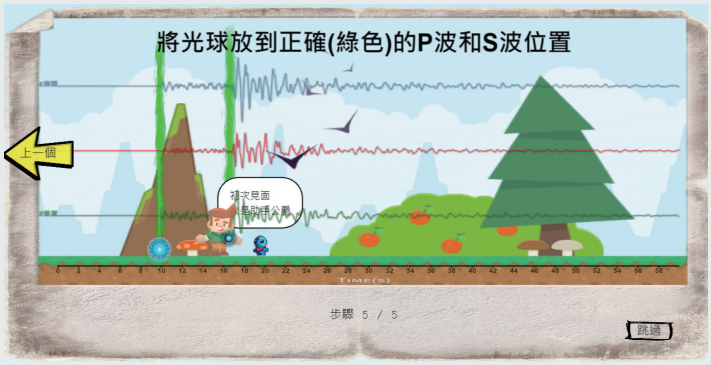


圖五、台灣地區的三維地震分布紙模型。

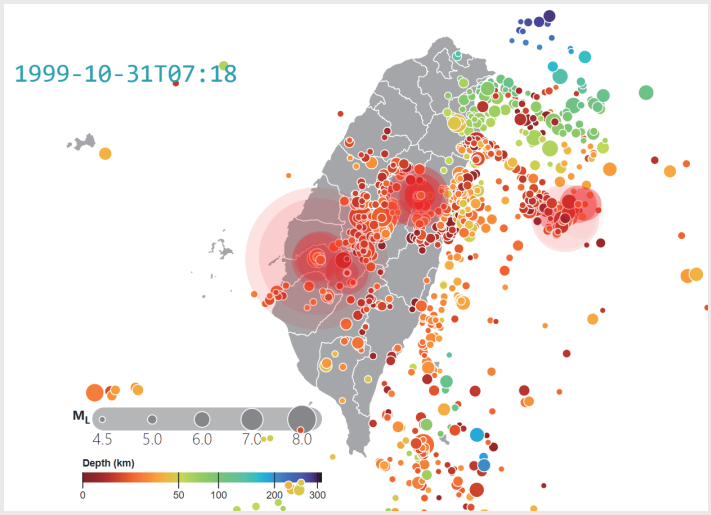
例如最近完成的網頁角色扮演遊戲 - 勇闖震源城，能讓玩家輕鬆認識地震波且根據自己的觀測決定地震的震央位置，並進一步探索震源以擊敗魔王拯救世界 (Chiu et al., 2022) (圖六)。

視覺化服務

為了協助使用者從事地震觀測的測站部署作業並與其他資訊套疊，TECDC 製作測站資訊轉 KML 格式的線上轉換工具，可以運用 Google Earth 強大的網路地理資訊系統功能進行空間分析；另一常用的線上繪圖應用是輸入地震儀器的零極值 (poles and zeros) 儀器響應來產出儀器的振幅與相位頻譜，藉以理解儀器的特性。在教育推廣的部分特別開發台灣 1991 年以來 ML>4.5 的地震時空分布，可由使用者調整播放的進度和速度 (圖七)。



圖六、利用網頁的角色扮演遊戲來認識地震波與地震定位的關聯。



圖七、台灣地震活動的時空分布播放介面。

結語

台灣地震科學資料中心的主要任務是支援研究人員的地震資料需求，本團隊持續鑽研資料管理的相關技術，近幾年一方面陸續整合既有的波形資料庫，並與各學研單位密切合作，針對有意開放的資料開發符合國際標準的網路服務系統，另一方面也積極打造研究成果的展示平台，擴大地震科學知識的應用價值。未來仍將稟持這樣的步調為全體學界做出貢獻。

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Wu, Y.-M., D.-Y. Chen, T.-L. Lin, C.-Y. Hsieh, T.-L. Chin, W.-Y. Chang, W.-S. Li, and S.-H. Ker (2013). A High-Density Seismic Network for Earthquake Early Warning in Taiwan Based on Low Cost Sensors. Seismological Research Letters, 84(6), 1048–1054. https://doi.org/10.1785/0220130085

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2022.06
台灣地震科學中心簡訊
Taiwan Earthquake Research Center Newsletter

The Taiwan Earthquake Research Data Center (TECDC)

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doi: 10.30067/TECNL.202206_1(1).0002

ABSTRACT

The Taiwan Earthquake Research Data Center (TECDC) is primarily sponsored by the Ministry of Science and Technology (MOST) and its aim is to integrate all available seismological data to serve the entire Taiwan research community. The TECDC is dedicated to the development of web services for the data sharing of both existing, backbone seismic networks and research-based temporary seismic arrays, and also for the demonstration of research outcomes and data products derived from domestic scientific outputs. We have also designed various online activities to promote seismological education and outreach. The official web portal of the TECDC is accessible at https://tecdc.earth.sinica.edu.tw (Fig. 1).

DATA INTEGRATION

To meet the demands of modern observational seismology developed in Taiwan, we have taken advantage of new web technologies to establish a web service system for the Broadband Array in Taiwan for Seismology (BATS). BATS is operated by the Institute of Earth Sciences, Academia Sinica and plays the role of the backbone seismic network in the Taiwan region. All BATS stations are equipped with state-of-the-art seismometers and the high-quality data collected have been open to the public since 1996 (Institute of Earth Sciences, Academia Sinica, Taiwan, 1996). The links of the web service system are available at http://bats.earth.sinica.edu.tw.

According to the Taiwan Earthquake Research Center (TEC) data delivery policy, Principal Investigators (PIs) who have utilized TEC equipment have to deliver their collected data and associated metadata to the TECDC. The delivered dataset will be archived at the TECDC and be accessible for all Taiwan-based researchers for a two year period after the seismic experiment. To integrate all the archived datasets, we train each research group on how to construct a standard database in which both daily

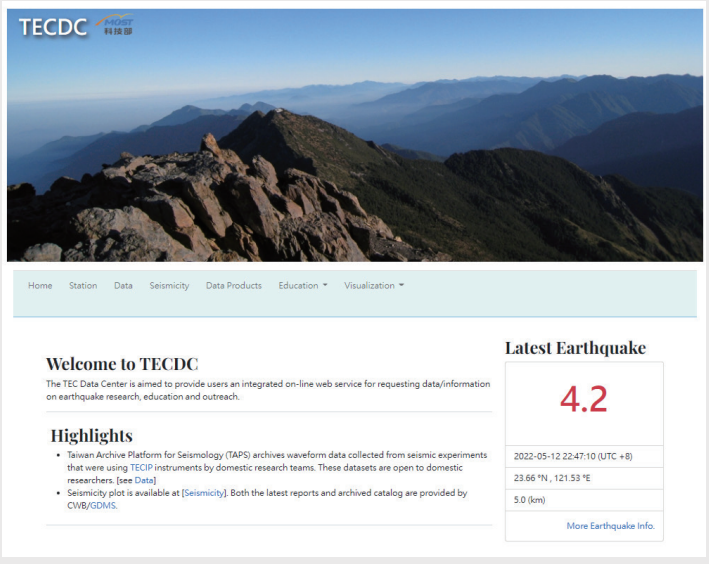


Figure 1. The web portal of the Taiwan Earthquake Research Data Center (TECDC) at https://tecdc.earth.sinica.edu.tw.

waveforms and associated metadata are organized in an appropriate manner. We also constructed the Taiwan Archive Platform for Seismology (TAPS) web services to enable the sharing of all experimental data under the TEC data policy. Users are able to request any data collected from the archived seismic experiments using just one platform, simply by registering an account. The web portal of TAPS is ready at https://taps.earth.sinica.edu.tw.

We have gained a lot of experience at creating seismic data web service systems over the past decade. The Taiwan Central Weather Bureau (CWB) intends to open up their data, which has been collected by various seismic networks and geophysical networks, through a collaboration between the CWB, IES and TEC beginning in 2020. We have successfully developed the Taiwan Seismological and Geophysical Data Management System (GDMS) to share these invaluable datasets with the world in 2022. The official website of the Taiwan GDMS is accessible at https://gdmsn.cwb.gov.tw.

Figure 2 shows the distribution of seismic stations operated by various organizations in Southwestern Taiwan. This online station map can be found at the TECDC.

REAL-TIME SEISMOLOGY

The TECDC has been collaborating with several research groups to practically develop Taiwan real-time seismology. For instance, the real-time intensity map, animation and event-based database of the P-Alert Strong Motion Network (Wu et al., 2013) (https://palert.earth.sinica.edu.tw). With regard to the near real-time source inversion of the BATS waveform data, we have not only collaborated with Dr. Shiann-Jong Lee to develop the Real-time Moment Tensor Monitoring System (RMT, Lee et al., 2013; http://rmt.earth.sinica.edu.tw) but have also automated the previous BATS CMT source inversion scheme (AutoBATS solution, Jian et al., 2018; https://tecdc.earth.sinica.edu.tw/FM/AutoBATS/) (Fig. 3). These solutions will be published in the Taiwan Earthquake Science Information System

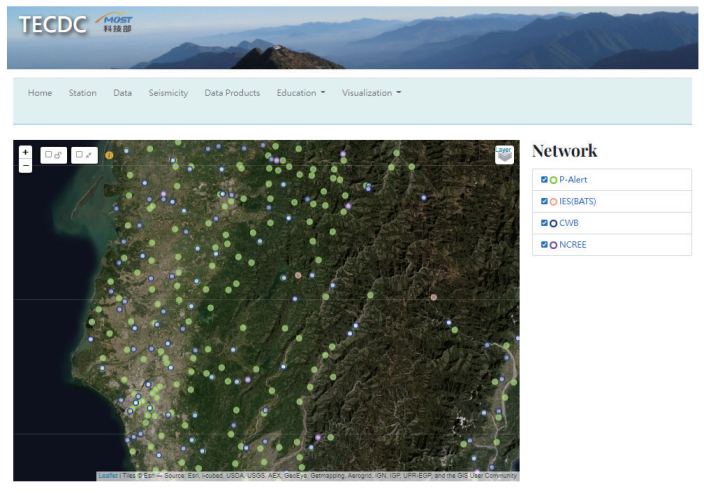


Figure 2. Distribution of seismic stations operated by various institutions in SW Taiwan.

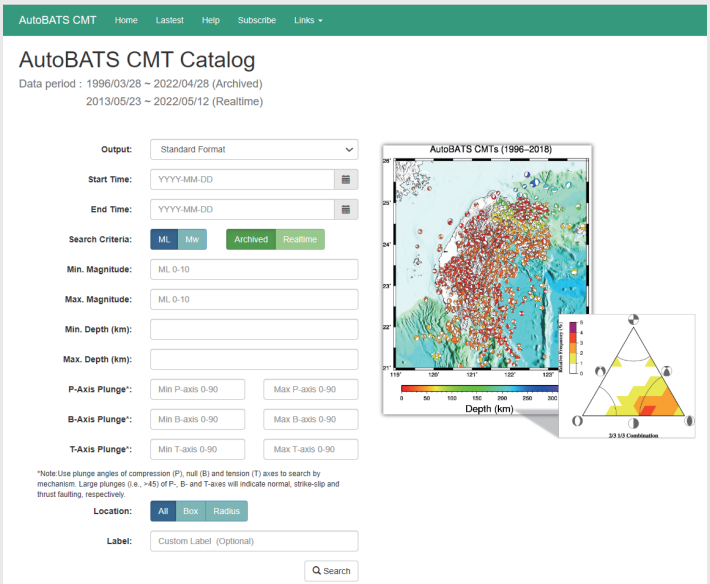


Figure 3. The homepage of the AutoBATS MT solution database.

(TESIS, https://tesis.earth.sinica.edu.tw) within five minutes after the occurrence of an earthquake and can be disseminated to subscribers after manual verification via e-mail. Furthermore, we performed the regional W-Phase source inversion technique (Kanamori & Rivera, 2008) to obtain a robust estimate of the magnitude and source mechanism of large earthquakes that occurred in the Taiwan area. To make sure the point source assumption was still valid, we exchanged real-time waveform data with neighboring countries to form a large aperture seismic network on a regional scale. The near real-time regional W-Phase solutions are presented at https://tecdc.earth.sinica.edu.tw/FM/WP/.

DATA PRODUCT DEPOSITORY

Referring to the IRIS Earth Model Collaboration (EMC; IRISDMC, 2011) product, we have constructed an online web service system for data product management and model previewing as the outcome of the Taiwan Velocity Model (TVM) project in order to provide the research community with access to various Taiwan 3D velocity and attenuation models. This is the first scientific outcome that has been hosted by the TECDC. This web service has archived 3D models that were published in the past three decades and the most updated model (Chen et al., 2019) will be accommodated within a short time. Users are able to select a model and mark a section on the map to plot a 2D profile of regional velocity or attenuation structure, which can be superimposed with seismicity and focal mechanisms. The TVM web service (Fig. 4) is accessible at https://tecdc.earth.sinica.edu.tw/TWtomo/.

In addition, we have designed interfaces for the visualization of Taiwan seismicity in several different ways. The Taiwan Seismicity Map is provided for the rapid assessment of the spatiotemporal distribution of aftershocks and historic earthquake sequences. On the other hand, the animation and 3D paper model are mainly for educational purposes. All these web services are available in the Seismicity function of the TECDC.

EDUCATION AND OUTREACH

The TECDC has developed an abundance of classroom activities for promoting self-learning ability in earthquake science (Liang et al., 2016). Users are able to use this facility to independently determine the epicenter and magnitude of earthquakes. We have provided a design for instructors and students to build an inertial

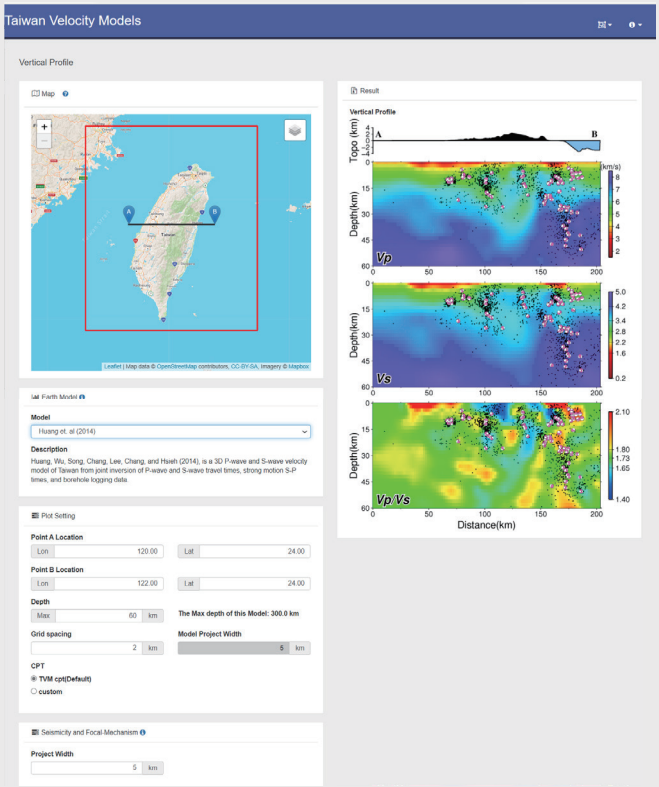


Figure 4. Web service of the Taiwan Velocity Model.

seismometer with low-cost materials (TEC-100), which demonstrates the basic physics behind a modern seismometer. We have also designed a handy paper model that shows the 3D distribution of the hypocenter that occurred in the Taiwan region (Fig. 5) to provide a better understanding of the tectonic setting of Taiwan through earthquakes. A recent development of the web-based role-playing game (RPG) about earthquake locations was published this June (Chiu et al., 2022), and we have added many additional game elements to make it more attractive (Fig. 6). Players are able to determine the epicenter easily by selecting the P and S waves for each seismic station. With the knowledge gained in the game, they will be equipped with the power to find the hypocenter, destroy the monster and save the world.

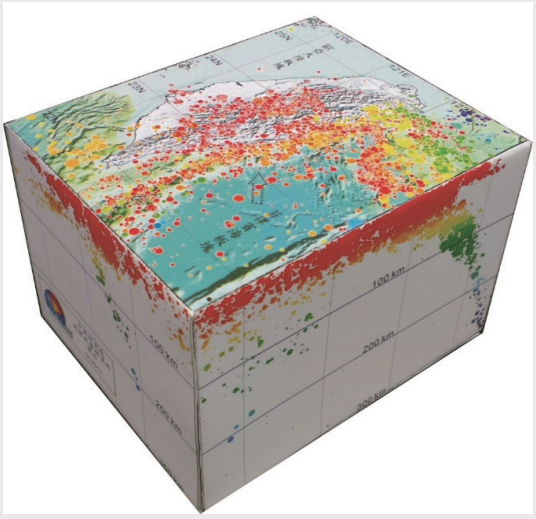


Figure 5. A paper model showing the 3D distribution of Taiwan seismicity.

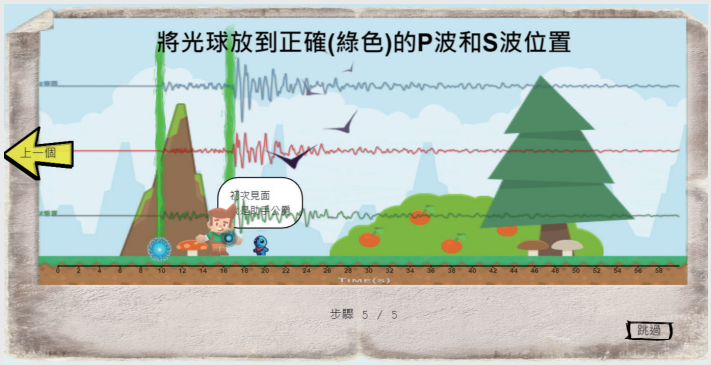


Figure 6. A web-based role-playing game about earthquake locations.

VISUALIZATION SERVICES

To help PIs plan the deployment of seismic arrays, we have provided an on-line converter that converts text content into a keyhole markup language (KML) file, which can be used to superimpose the station information onto Google Earth, a powerful web-GIS platform supported by Google. Another service plots the instrument response, in which both the amplitude and phase spectra are presented from the pole-and-zero values input by users. This is essential for understanding the characteristics of the seismograph typically used in practice. Figure 7 shows a snapshot of an online Taiwan seismicity animation accumulated from earthquake data from 1991-01-01 to 1999-10-31 with ML > 4.5. Both the progress and the play speed can be adjusted by users.

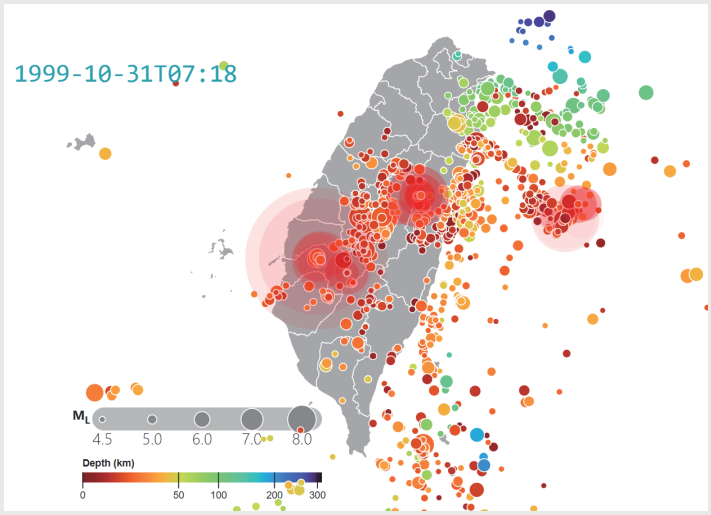


Figure 7. A snapshot of a Taiwan seismicity animation accumulated for earthquakes with ML > 4.5 from 1991 to 1999-10-31.

CONCLUDING REMARKS

The main mission of the TECDC is to support the development of infrastructure, which aims to open up seismological data to researchers. We have continuously accumulated experience over the past two decades in modern seismic data technology. In recent years, we have integrated existing seismic waveform databases and collaborated with various research institutions to develop web service systems for data sharing in a standard format. To broaden the possible applications of achieved scientific outcomes, we are developing a data products depository for the scientific results derived by Taiwanese geoscientists. We will definitely continue this work, hoping to increasingly contribute to the entire research community.

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台灣地震科學中心簡訊
Taiwan Earthquake Research Center Newsletter

台灣地震科學中心地震儀器服務平台

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doi: 10.30067/TECNL.202206_1(1).0004

摘要

台灣地震科學研究中心深入探討台灣地震之孕震、發震、震裂之科學過程，積極推動許多與地震相關的研究計畫，諸如台灣地震模型、地震預警與前兆研究及台北盆地之強震放大效應等，為因應各項研究子計畫對儀器之需求，科技部自然司地球科學學門已於 2006 年成立台灣地震科學中心地震儀器服務平台，以共同使用、集中管理、提高地科儀器之使用效率及維護為目標。此一服務平台建置於中央研究院地球科學研究所，設有共用儀器及維修體系，提供受過專業訓練的技術人力，協助各研究計畫有關儀器建議、安裝、及資料初步處理之工作。本平台並設有儀器管理委員會，制定儀器使用規則、審核儀器申請與監督平台執行，委員由台灣地震科學中心執行委員會推薦相關大學和研究機構之學者專家組成。

表一、台灣地震科學中心地震儀器服務平台之儀器 (統計至 2021 年 12 月)。

分類	儀器名稱	說明	數量	總數量
GNSS	RS	GNSS receiver	12	108
	R7	GNSS receiver	6	
	R8	GNSS receiver	11	
	R9	GNSS receiver	19	
	U-blox F9 Receiver	GNSS receiver	60	
數位地震儀 (地震儀 + 紀錄器)	CMG-5TD	ccelerometer	10	52
	CMT-3ESPCDE	broadband sensor	12	
	Meridian Compact	broadband sensor	30	
記錄器	Q330	6-channel datalogger	19	114
	Q330S	6-channel datalogger	10	
	Q330S+	6-channel datalogger	10	
	Centaur	6-channel datalogger	4	
	Data Cube	3-channel datalogger	71	
地震儀	Trillium 120	broadband sensor	14	110
	Trillium compact	broadband sensor	19	
	CMG-3ESPC	broadband sensor	12	
	CMG-40T	broadband sensor	4	
	Geophone (2 Hz, 3-axis)	short-period sensor	6	
	Geophone (4.5 Hz, 3-axis)	short-period sensor	55	
密集地震儀陣列儀器	SmartSolo	三分量短周期地震儀	100	100
尋北儀	Quadrans	fiber-optic gyrocompass	1	1
VSAT 衛星傳輸設備	Carina	小型衛星資料傳輸 (中心站)	2	5
	Cygnus	小型衛星資料傳輸 (野外站)	3	
地震資料處理軟體	Antelope	地震資料處理	1	1
手持式頻譜分析儀	R&S FSH4	VSAT 系統架設與測試	1	1
直趨式旋轉運動模擬器	Aerotech ARMS200	檢驗旋轉感測器	1	1

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平台儀器

師法美國地震學研究聯合會 IRIS 運作 PASSCAL 儀器中心的成功經驗，國科會 (現科技部) 在 2006 年成立台灣地震科學中心 (以下簡稱中心) 地震儀器服務平台 (以下簡稱平台)，每年陸續採購各式儀器設備，至 2021 年 12 月為止，平台現有的儀器設備計有 GNSS 精密衛星定位儀設備 108 組、地震儀 110 組 (寬頻 49 組、短周期 61 組)、地震記錄器 114 組、數位地震儀 52 組、密集陣列地震儀 100 組、尋北儀系統 1 組、小型衛星通信系統及資料處理軟體等。

儀器借用制度

平台所採購之地震相關儀器需使用於研究與教育用途，平台設有儀器管理委員會，制定儀器使用規則、審核儀器申請與監督平台執行，委員由中心執行委員會推薦相關大學和研究機構之學者專家組成，根據『台灣地震科學中心地科儀器維護與服務平台』設置及管理辦法，儀器申請期間為每年 7 月底為止，經儀器委員會討論，最晚於當年 10 月底公告審查結果，視儀器使用規劃及儀器數量借出。儀器借用以兩年為限，重大計畫需長期借用者，須提出詳細規劃，經儀器管理委員決議討論通過始得借用。若儀器有庫存，在申請期間之外可隨時接受儀器申請，並視儀器使用規劃借出，惟借用期限以不超過兩年為限。利用平台設備所蒐集之資料需在實驗結束兩年後透過台灣地震科學資料中心所開發之台灣地震科學資料服務平台公開。

為有效利用儀器須凝聚整體社群之共識，計畫主持人在規劃實驗階段有儀器上的任何問題都歡迎與平台管理人員討論，透過公開交流溝通各種情況如實驗環境較合適之儀器種類、儀器借用規劃是否與他人有所衝突等都能即早協商，以利備案討論。

為方便儀器申請、儀器介紹、平台管理與資料共享，平台建置三個網站：

- 儀器申請網站 https://tec.earth.sinica.edu.tw/instrumentapply/
- 儀器介紹網站 https://www.elab.earth.sinica.edu.tw/tecip
- 台灣地震科學資料服務平台網站 https://taps.earth.sinica.edu.tw/zh-TW/。

經由儀器申請網站向平台申請儀器時，需注意以下事項：

- 簡短描述實驗如何進行，包含會遭遇之特殊野外情況
- 實驗的位置 (經緯度與區域)
- 實驗地日期，包含儀器預計借出與收回的時間，註明不可抗拒因素之下儀器可能延後歸還之情況



圖一、平台儀器申請網站。



圖二、平台儀器介紹。



圖三、台灣地震科學資料服務平台。

- 實驗所需儀器的種類與數量
- 預期可蒐集實驗資料之時間區段
- 註明任何需要特別協助之處
- 提供經費來源與經費狀況
- 實驗聯絡人之聯絡資訊 (email、電話、地址)

儀器借用須配合繳交兩份報告，儀器安裝完成後兩個月內繳交安裝報告，以及實驗結束後三個月內繳交實驗報告，報告內容以交代儀器安裝經緯度、安裝方式 (照片)，測站環境描述，歷次維護紀錄，透過報告可讓不熟悉實驗之資料使用者能充分地理解實驗內容，以便決定是否使用該資料。報告並無制式的格式但內容須涵蓋：

- 實驗名稱及簡稱
- 進行實驗之 PI 的姓名與單位
- 詳細描述實驗的目的
- 表格列出測站位置，以及說明資料蒐集情況
- 說明實驗的設計與執行情況
- 詳細說明實驗進行期間所遭遇之不尋常事件 (如更換設站位置、記錄器失去 GPS 訊號等)
- 說明實驗期間遭遇的困難，包含硬體故障或受損
- 任何建議讓平台更加進步
- 地圖或其他圖表 (非必要)

資料繳交制度

繳交平台儀器所蒐集之資料是 PI 的責任與義務，資料的公開將有利與全體社群。實驗完成後，儀器歸還時須繳交儀器所獲得的全部原始觀測資料，若未達成承諾，將依情節輕重停止該計畫主持人儀器借用權利一至三年。在實驗完成後兩年內，執行計畫之 PI 可透過資料中心網站下載整理後之資料，PI 也可放棄此權力並提前公開資料；實驗完成兩年後，資料對台灣地震科學中心之成員公開。

當平台之儀器使用於教育教學用途而非科學觀測的情況下，向平台管理人員申請並獲得同意下，該資料可豁免公開；其他資料可豁免公開的情況，包含實驗失敗而無法獲取資料，或該資料不含有意義的數據。

參考資料

The Incorporated Research Institutions for Seismology (IRIS) Portable Array Seismic Studies of the Continental Lithosphere (PASSCAL) Instrument Center. https://www.passcal.nmt.edu/

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台灣地震科學中心簡訊
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Taiwan Earthquake Research Center Instrument Service

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Institute of Earth Sciences, Academia Sinica
doi: 10.30067/TECNL.202206_1(1).0004

ABSTRACT

The Taiwan Earthquake Research Center studies seismogenic and rupture processes, engaging in precise monitoring and measurement of deformation across the entire island of Taiwan. The center also acts as a platform for advanced research e.g., the Taiwan Earthquake Model, Earthquake Early Warning, Earthquake Precursor, and the ground motion simulation of the metropolitan city. To provide technical support for this research, an instrument center supported by the National Science Council (now the Ministry of Science and Technology) was established in 2006 and is operated by the Institute of Earth Sciences, Academia Sinica. The instrument center contains several state-of-the-art portable broadband seismographs, continuous global navigation satellite system tracking equipment, and advanced data processing and database management tools to assist scientists in acquiring and archiving seismic information. Furthermore, the facility provides experts for consulting, installation, data archival and other services. An Instrument Advisory Committee made up of impartial members from the TEC Executive Committee along with any interested university representatives is designed to make policies, approve requests, and maintain community oversight.

INVENTORY

To follow the model of the Incorporated Research Institutions for Seismology (IRIS) Portable Array Seismic Studies of the Continental Lithosphere (PASSCAL) Instrument Center, the world's largest academic pool of seismological research instrumentation, the Taiwan Earthquake Research Center Instrument Pool (TECIP) funded by the Department of Natural Sciences and Sustainable Development, National Science Council (now the Ministry of Science and Technology) was established in 2006. The TECIP continues to expand its inventory every year. As of 2021/12, the inventory in service consists of 108 continuous global navigation satellite system (GNSS) tracking instruments, 114 seismic data loggers, 49 broadband seismometers, 61 short-period geophones, 52 digital seismometers (data loggers and sensors), 100 dense array seismometers, one gyrocompass, a very small aperture terminal (VSAT) transmission system, Antelope data processing software, and a calibration facility (Table 1).

INSTRUMENT USE POLICY

The seismic instruments purchased by the TECIP are available to any research or

Table 1. TECIP instrument inventory (as of 2021/12).

Category	Instrument	Description	Quantity	Total
GNSS	RS	GNSS receiver	12	108
	R7	GNSS receiver	6	
	R8	GNSS receiver	11	
	R9	GNSS receiver	19	
	U-blox F9 Receiver	GNSS receiver	60	
Digital seismometer (data logger + sensor)	CMG-5TD	Accelerometer	10	52
	CMT-3ESPCDE	Broadband sensor	12	
	Meridian Compact	Broadband sensor	30	
Data logger	Q330	6-channel datalogger	20	115
	Q330S	6-channel datalogger	10	
	Q330S+	6-channel datalogger	10	
	Centaur	6-channel datalogger	4	
	Data Cube	3-channel datalogger	71	
Seismometer	Trillium 120	Broadband sensor	14	110
	Trillium compact	Broadband sensor	19	
	CMG-3ESPC	Broadband sensor	12	
	CMG-40T	Broadband sensor	4	
	Geophone (2 Hz, 3-axis)	Short-period sensor	6	
	Geophone (4.5 Hz, 3-axis)	Short-period sensor	55	
Dense array seismometer	SmartSolo	Node system	100	100
North finder	Quadrans	Fiber-optic gyrocompass	1	1
VSAT transmission system	Carina	Central station	2	5
	Cygnus	Field station	3	
Data processing software	Antelope	Seismic data processing	1	1
Handheld spectrum	R&S FSH4	VSAT signal spectrum	1	1
Rotation simulator	Aerotech ARMS200	Rotation shake table	1	1

educational institution to use for research purposes within the guidelines established in this document. An Instrument Advisory Committee of impartial members from the TEC Executive Committee along with any interested university representatives is designed to make policies, approve requests, and maintain community oversight. According to the TECIP regulations, the duration of an instrument loan for an experiment is limited to two years. Special requests for longer duration of an instrument loan is possible and need to be discussed by TEC Executive Committee. The TEC Data Center (TECDC) will make the data available only to the principal investigator (PI) or designated representative(s) for a period of up to two years after the completion of the experiment. The instrument request needs to be submitted by no later than the end of July. The TEC will publish the schedule no later than the end of October as soon as the committee recommendations are completed. Once the experiment has been scheduled, the PI will be contacted to work out the details about the exact type of equipment requested. Subject to equipment availability (so as not to interfere with previously approved experiments), instrument requests can also be made at any time during the year, and they will be made available to users as the schedule permits.

The efficient use of the instruments will require close cooperation among all of the parties involved. The principal investigator is encouraged

to contact the TECIP manager about any planned experiment during the proposal development stage in order to determine if there would be any problems in operating the equipment in the environment required for the experiment. It is also important for everyone to know of possible scheduling conflicts as early as possible. Open communications will allow the development of alternative plans early in the scheduling process.

The TECIP has set up three websites: one for online instrument requests https://tec.earth.sinica.edu.tw/instrumentapply/, one to introduce a variety of instruments https://www.elab.earth.sinica.edu.tw/tecip, and a data archive at the Taiwan Archive Platform for Seismology (TAPS), developed and managed by Taiwan Earthquake Research Data Center (TECDC) https://taps.earth.sinica.edu.tw/zh-TW/. (See Figures 1 to 3.)

The initial request should be submitted to the TEC via forms on the TECIP website and must include the following information:

- A short description of the experiment to be conducted, including any unusual field conditions that may be encountered;
- The location of the experiment (latitude and longitude, as well as an estimate of the aerial extent);
- Dates that the equipment will be shipped from and returned to the TECIP, including additional information on any circumstances that may make it impossible to shift the dates forward or backward;
- The types and number of pieces of equipment requested for the experiment;
- An estimate of the amount of data to be gathered and archived;
- A notification of any special support that may



Figure 1. TECIP instrument request website.



Figure 2. TECIP instrument introduction website.



Figure 3. TAPS data archive website.

be required;

- The name of the funding agency and status of the funding support; and
- A mailing address, email address, and telephone and fax numbers for the designated contact person for this experiment.

All PIs are required to submit two reports for each instrument request. The Installation Report must be submitted via the instrument request website within two months after the fieldwork has been completed. The Experiment Report must be submitted via the instrument request website within 3 months after the demobilization date. The Installation Report summarizes the field portion of the experiment. The Experiment Report should contain enough information that a researcher who is unfamiliar with the project can scan the report and determine whether or not the data would be useful to them. Although specific requirements for the format of the report do not exist, all reports submitted should include:

- The name and institutions of the Principal Investigators;
- The experiment name and abbreviation;
- A paragraph describing the purpose of the experiment;
- A description of the layout and execution of the experiment;
- Details of any unusual situations encountered during project mobilization (changing locations, missing GPS timing, etc.);
- A description of any problems encountered with either the hardware or the software furnished by the TECIP program;
- Recommendations for further improvements of the facility; and
- Maps and other figures, which are welcomed and encouraged but not required.

DATA DELIVERY POLICY

The delivery of data collected by the PI using the TEC equipment and associated metadata to the TECDC is an obligation of the PI. Abiding by this data policy ensures the continued contribution of the TECIP facility to seismological studies. Failure to complete this requirement not only deprives the community of a valuable data resource, but also may jeopardize future requests to borrow TEC equipment. The TECDC will make the data available only to the PIs and their designated representative(s) for a period of up to two years after the completion of the experiment. The PI can waive this period of exclusive access. After the prescribed moratorium, the data will be made freely and openly available to TEC members.

If you are using equipment from the TECIP facility for strictly educational training on the instrumentation and are not using the equipment to pursue a scientific target, TECIP management may grant an exemption from the data distribution policy if there is no useful information to be shared among the seismological community. There may be other needs for exemptions, including complete failure of the experiment to capture data or failure to capture useable data, and these will be granted on an as-needed basis.

REFERENCE

The Incorporated Research Institutions for Seismology (IRIS) Portable Array Seismic Studies of the Continental Lithosphere (PASSCAL) Instrument Center. https://www.passcal.nmt.edu/

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台灣地震科學中心簡訊
Taiwan Earthquake Research Center Newsletter

台灣地震科學中心地球物理儀器服務平台

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在台灣地震科學中心之架構下，以「地震儀器服務中心」為藍圖，由中央大學地球科學系負責，在 2011 年成立「地球物理儀器服務中心」，發展地震儀器以外的地球物理方法所需儀器，主要包括震測、重磁、地電、地熱、海上地球物理、及井測等事項，及資料處理服務平台。本計畫結合原地震儀器服務中心，建立及擴大儀器服務架構，提供基本地球物理研究儀器，服務地科學界。目前已建置完成之儀器池種類繁多，如海底地震儀、井測、MT、透地雷達、地電阻、重力、磁力、震盪震源車及震測收錄設備等（如下表），細細資訊請參考 TEC 網站服務平台之儀器中心介紹。

建置儀器共用系統，須設立專責管理機構，有效整合並管理儀器資源，始能發揮其最大的科學價值。儀器中心除了儀器方面的專責工作人員外，並設有儀器委員會，負責協調科學家對新儀器的需求，決定儀器使用規則並審核儀器的規格。儀器委員會除提供縱向的監督管理並審核年度的使用計畫外，在橫向工作方面，需與台灣地震研究中心各研究計畫有密切的聯繫與合作，並要求各計畫於實驗工作結束後，於一定期限內繳交統一格式的觀測資料，開放給其他研究者共同使用。

本計畫亦針對各種地球物理儀器，設計專業實習課程，提供國內各大學地球物理相關課程之實習支援，透過各種地球物理儀器的實際接觸與操作，有助於學生對於儀器的探測方式加深實務印象與增加學習的興趣。申請方式請至 TEC 網站：https://tec.earth.sinica.edu.tw/geoclass/

類別	儀器名稱	數量
震測	震盪震源車（EnviroVibe）	1
	24 波道震測資料收集模組（Geode）	7
	資料採集系統（SmartSolo PDH、PBC）	1
	三分量地震（SmartSolo 3C 5Hz）	60
	單分量地震儀（SmartSolo 1C 10Hz）	315
重磁	相對型重力測勘儀（SCINTREX CG-6）	1
	全磁場磁力儀（GSM-19GW v7.0）	3
	三分量磁力儀（MAG-3ADL2 +MAG-03TB）	2
	三分量磁力儀（FRG-604RC）	1
	即時動態衛星定位系統（Trimble R8s LT）	1
地電	超低頻衛星同步電磁波探測系統（MTU-5A）	2
	遠端地電阻長期監測及探測系統（SeeHope R2）	4
	透地雷達（pluse EKKO-PRO）	2
井測	井測系統（RG Micrologger II）	1
海球物理	海底地震儀（MicrOBS+）	14

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台灣地震科學中心簡訊
Taiwan Earthquake Research Center Newsletter

The Geophysical Instrument Service

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The Geophysical Instrument Service center was established in 2011 under the framework of the Taiwan Earthquake Research Center and using the "Seismic Instrument Service Center" as the blueprint. The Department of Earth Sciences of the Central University is in charge of the center, which was established to develop the instruments required for the implementation of geophysical methods, including seismic, gravity, magnetic, and geoelectric surveys, geothermal processes, marine geophysics, and well logging, as well as the provision of data processing service platforms. Combined with the Seismic Instrument Service Center, this plan establishes and expands the instrument service structure, provides basic geophysical instruments, and serves the geoscience community. Many types of instruments have been deployed, such as ocean bottom seismometers, well logging equipment, magneto-telluric equipment, ground-penetrating radar systems, electrical resistivity equipment, gravimeters and magnetometers, vibrator vehicles, and seismic recording equipment (Table 1). For detailed information, please refer to the TEC website.

To build an instrument sharing system, it is necessary to set up a management agency to effectively integrate and manage instrument resources so that the maximum scientific value can be extracted. In addition to instrument technicians, the Instrument Center also has a committee that is responsible for coordinating the demand for new instruments, determining instrument usage rules, and reviewing instrument specifications. In addition to providing supervision, management, and the reviewing of the annual use plan, the Committee needs to have close contact and cooperation with the research projects of the Taiwan Earthquake Research Center. The submission of observational data in a unified format within a certain period of time and availability to other researchers for shared use is another important consideration.

This project also designs internship courses for various geophysical instruments and provides applications for geophysics-related courses in universities. Through the practical operation of geophysical instruments, it helps to increase student interest in learning about the detection methods of instruments. Please visit the TEC website to apply: https://tec.earth.sinica.edu.tw/geoclass/

Category	Instrument Name	Quantity
Seismic Survey	EnviroVibe	1
	Geode (24 channels)	7
	SmartSolo: PDH, PBC	1
	SmartSolo 3C 5Hz	60
	SmartSolo 1C 10Hz	315
Gravity and Magnetic	SCINTREX CG-6	1
	GSM-19GW v7.0	3
	MAG-3ADL2 +MAG-03TB	2
	FRG-604RC	1
	Trimble R8s LT	1
Geoelectricity	Phoenix MTU-5A	2
	SeeHope R2 (64 channels)	4
	pulse EKKO-PRO GPS	2
Well Logging	RG Micrologger II	1
Marine Geophysics	Sercel MicrOBS+	14

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